

SHAH INVESTOR'S HOME LIMITED

Issue Tracker & Service Request System

Business Requirements Specification — a module of Synapse

Document	Business Requirements Specification (BRS)
System	Issue Tracker & Service Request System, embedded within SIHL Synapse
Organisation	Shah Investor's Home Limited (SIHL), SEBI-registered stockbroker
Version	1.0
Status	Draft — Phase 1 ready for build; Phase 2/3 directional
Audience	Management, business stakeholders, project sponsors

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1. Purpose and Scope

1.1 Purpose

SIHL currently handles issues and service requests informally — phone calls into RMS, ad-hoc WhatsApp threads, undocumented email chases — with no single record of what was raised, by whom, or how long it took to resolve. This has already produced avoidable outcomes, including a client penalty caused by a verbal instruction being misheard over a phone call to RMS.

This module gives SIHL a **single ticketing and service-request system**, embedded in Synapse, covering every channel through which an issue can be raised, with a complete audit trail from intake to resolution.

1.2 In scope — Phase 1

- Ticket intake across Website, WhatsApp, Phone and the Synapse application itself.
- Automatic identification of the raiser against existing Synapse master data, with a fallback path for people not yet on record.
- A predefined, business-owned issue taxonomy used for both routing and reporting.
- Detection and merge of duplicate tickets, confirmed by a human before anything is combined.
- A resolution workflow with forwarding, reminders and a complete audit trail.
- Conversion of larger requirements into change requests, gated by management approval.
- Automated delivery of routine documents (ledgers, margin reports, contract notes) as a lighter-weight service-request flow.
- A basic operational stats report.

1.3 Out of scope / explicitly deferred

- **Management dashboard** (CEO Command Center integration) — Phase 2.
- **Automated email intake** via Google APIs — Phase 3.
- **Solutions repository and after-hours chatbot** — Phase 3.
- **OTP/MFA on document requests** — considered and explicitly not included; email/mobile match against master data is sufficient for Phase 1.
- **Severity-tiered response SLAs** — considered and explicitly not included; Phase 1 uses one flat response target for every ticket.

2. Business Context

SIHL's support load spans internal employees, B2B customers, associate partners, clients and prospects, arriving through whichever channel is most convenient for the person raising the issue. Without a shared system of record, three problems recur:

- **No accountability trail.** When an issue is handled by phone or informal chat, there is no record of who decided what, which becomes a real liability when a decision turns out to be wrong.
- **Duplicated effort.** The same issue is often raised more than once across different channels, with no mechanism to recognise and merge it.
- **Manual, repetitive service work.** Routine document requests (ledgers, margin reports, contract notes) consume staff time that a simple automated flow can remove entirely.

This module addresses all three by giving every issue a single, trackable record from the moment it is raised, and by automating the service requests that do not need human judgement at all.

3. Stakeholders and User Roles

Actor	Role in the system
Employee	Raises internal issues; may raise a request on behalf of a client
B2B customer / Client	Raises issues and service requests; already present in Synapse master data (hierarchy code 0800)
Associate partner / Prospect	Raises issues; typically absent from master data — handled via the guest ticket path
Support desk staff	Triages, resolves or forwards tickets
Support lead / desk head	Owns and revises the issue-category taxonomy
Product / Ops management	Approves conversion of tickets into change requests

4. Business Objectives

Ref	Objective	Success indicator
OBJ-1	Give every issue a single, trackable record regardless of the channel it arrived on	100% of issues raised via Website, WhatsApp, Phone or Synapse produce a ticket
OBJ-2	Remove ambiguity about who is accountable for an issue at any point in time	Every forward and status change is attributed and timestamped
OBJ-3	Eliminate duplicate handling of the same issue	Duplicate tickets are surfaced to staff before they are worked twice
OBJ-4	Remove manual effort from routine document requests	Ledger, margin report and contract note requests are delivered without staff intervention

OBJ-5	Give management visibility into support volume and turnaround	A basic stats report in Phase 1, a full dashboard in Phase 2
OBJ-6	Lay groundwork for after-hours self-service	Resolved-ticket history is structured well enough in Phase 1 to seed a curated solutions repository in Phase 3

5. Functional Requirements — Phase 1

5.1 Ticket intake

Ref	Requirement	Priority
FR-1.1	Tickets may be raised via Website (using predefined subject lines mapped to prescribed reasons), WhatsApp, Phone (logged by staff on the caller's behalf), and the Synapse application's own ticket-raise screen.	Must
FR-1.2	Email intake is not supported in Phase 1; automated email reading depends on the Google API integration deferred to Phase 3.	Must
FR-1.3	Every ticket records: raiser identity (or guest details), channel, category, description, timestamp and initial status.	Must

5.2 Identity resolution

Ref	Requirement	Priority
FR-2.1	On ticket creation, the system resolves the raiser against existing Synapse master data by email or mobile number.	Must
FR-2.2	If no match is found, the system creates a guest ticket, capturing name and contact details as given, without blocking ticket creation.	Must
FR-2.3	Guest tickets may be reconciled to a master data record later, for example once a prospect is onboarded as a client.	Should
FR-2.4	On creation, an automated registration notification (email or SMS) is sent confirming the ticket number and expected response window.	Must

5.3 Categorisation

Ref	Requirement	Priority
FR-3.1	A predefined issue taxonomy, structured as Product/Module × Issue Type, is used both to route tickets and as the reporting dimension.	Must
FR-3.2	The taxonomy is owned and revised by the support lead / desk head as new issue patterns emerge.	Must
FR-3.3	Category is selected at the point of intake, by the raiser on Website/Synapse or by staff when logging a Phone/WhatsApp ticket.	Must

5.4 Duplicate detection and merge

Ref	Requirement	Priority
FR-4.1	While a staff member is answering any ticket, the system surfaces other open tickets as likely duplicates using structured matching: same category, same client/entity, and raised within a rolling 24–48 hour window.	Must
FR-4.2	Matching is deterministic on structured fields in Phase 1; free-text similarity matching is not included.	Must
FR-4.3	Staff must manually confirm a duplicate via a checkbox before any merge takes place — the system never merges automatically.	Must
FR-4.4	Confirmed duplicates are combined into a single thread, carrying forward both tickets' history.	Must

Note. The combined thread's turnaround-time clock uses the earlier of the two original raised timestamps, so merging cannot be used to understate elapsed time in the average-turnaround metric (§8, FR-8.1). Pending final sign-off.

5.5 Resolution lifecycle

Ref	Requirement	Priority
FR-5.1	Support desk staff resolve a ticket directly or forward it to another team member.	Must
FR-5.2	Every forward is logged with who forwarded it, to whom, when and any accompanying remark, preserving a complete audit trail.	Must
FR-5.3	Staff receive a daily morning reminder listing their pending tickets and burn-down status.	Must
FR-5.4	Staff must update status and remarks on each ticket — marking it solved or providing a revised timeline.	Must
FR-5.5	The initial response target is a flat 12–24 hours for every ticket, regardless of category. Severity-based SLA tiers are explicitly out of scope for Phase 1.	Must

5.6 Change request conversion

Ref	Requirement	Priority
FR-6.1	An issue identified as a larger development requirement, rather than a fixable bug, may be converted into a change request.	Must
FR-6.2	Conversion requires approval from Product/Ops management; the support desk cannot park an item in the backlog unilaterally.	Must

FR-6.3	On conversion, the originating ticket is closed with a notification that it is being considered for a future release, and the item is parked as an enhancement in the application backlog.	Must
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5.7 Automated service requests

Ref	Requirement	Priority
FR-7.1	A lighter-weight flow, separate from the general ticket, handles automated document delivery: ledgers, margin reports and contract notes.	Must
FR-7.2	Ledger requests require a mandatory client code.	Must
FR-7.3	Verification for document delivery is by email/mobile match against master data; no additional OTP/MFA step is required in Phase 1.	Must
FR-7.4	The generated document is emailed directly to the client's registered email address.	Must
FR-7.5	If an employee raises the request on a client's behalf, the client is copied on the delivery for transparency.	Must

5.8 Reporting

Ref	Requirement	Priority
FR-8.1	A stats report tracks daily ticket closures, new tickets raised, tickets forwarded and average turnaround time.	Must
FR-8.2	No dashboard UI is included in Phase 1 — the CEO Command Center integration is Phase 2 (\$12).	Must

6. Business Rules

These govern behaviour that is not self-evident from the requirements above and must be preserved in any future change.

Ref	Rule
BR-1	A ticket must never fail to be created because the raiser cannot be identified — the guest ticket path is the fallback, not an error state.
BR-2	Tickets are never merged automatically; a human must confirm every merge.
BR-3	Change requests can only be created through management approval — never unilaterally by the support desk.
BR-4	Document service requests deliver only to the client's registered email address on file, never to an address supplied in the request itself.
BR-5	The issue-category taxonomy has one accountable owner (the support lead / desk head) at all times.

7. Non-Functional Requirements

Ref	Requirement
NFR-1	Ticketing data resides in the existing Synapse PostgreSQL database — no new datastore is introduced.
NFR-2	Every status change and forward is attributable to a specific user and timestamp, supporting both operational audit and any future dispute resolution.
NFR-3	Access control reuses Synapse's existing session and role model — no separate authentication system is introduced.
NFR-4	Identity resolution and guest-ticket fallback must not introduce a blocking failure mode on ticket creation.

8. Indicative Data Entities

High-level entities anticipated for Phase 1; detailed column-level design is a technical-architecture concern, not covered in this business specification.

Entity	Purpose
Ticket	The core record — raiser, channel, category, description, status, and merge linkage where applicable

Issue Category	The Product/Module × Issue Type taxonomy used for routing and reporting
Ticket Event	The audit log of forwards, status changes and remarks against a ticket
Service Request	Document-delivery requests, tracked separately from general tickets
Change Request	Items converted from a ticket and approved by Product/Ops management, linked back to their originating ticket

9. Assumptions, Dependencies and Constraints

9.1 Assumptions

- Employees and clients already exist in Synapse master data (the existing `userdetails` / `hierarchy` tables), which identity resolution can query directly.
- Associate partners and prospects are generally not present in master data, which is why the guest ticket path exists.
- The support lead / desk head is available and willing to own the issue-category taxonomy on an ongoing basis.

9.2 Dependencies

- This module depends on Synapse's existing authentication, session and master-data model — it is built as a module inside Synapse, not a standalone system.

9.3 Constraints

- Google API email intake and the after-hours bot are out of scope until Phase 3, and the bot specifically depends on a populated, curated solutions repository — it cannot be brought forward ahead of that content existing.

10. Roadmap and Future Phases

Phase	Contents	Depends on
Phase 1	Core ticketing loop: multi-channel intake, identity resolution with guest fallback, categorisation, duplicate merge, resolution with audit trail, change-request conversion, automated document service requests, basic stats report	Synapse's existing authentication and master data
Phase 2	CEO Command Center dashboard	Phase 1 data must exist and be reliable — this phase is purely additive reporting
Phase 3	Google API email intake and auto-categorisation; solutions repository; after-hours bot	Phase 1 resolution history feeds the repository; the repository must hold curated content before the bot has anything reliable to answer from

10.1 Phase 2 — CEO Command Center dashboard (directional)

Ref	Candidate capability	Note
FUT-1	Ticket aging shown in buckets: 6h / 24h / 48h / >48h	Builds on the turnaround data already captured in Phase 1
FUT-2	Product-wise issue count table, expandable per product to the specific topics driving volume	Surfaces systemic problems, not just raw counts
FUT-3	Exact dashboard KPI set	Not yet finalised — to be scoped at the start of Phase 2

10.2 Phase 3 — automation and self-service (directional)

Ref	Candidate capability	Note
FUT-4	Google API email intake, auto-categorising incoming email as problem vs. general query	Specific APIs not yet chosen — to be scoped at the start of this phase
FUT-5	Solutions repository built from resolved tickets	A resolution must be marked "reusable" by a staff member before entering the bot's answerable corpus — nothing is auto-published
FUT-6	After-hours bot on WhatsApp and Website chat	Answers strictly from the curated repository, given the compliance sensitivity of this domain; on no confident match it auto-creates a ticket for next-business-day human response rather than guessing

11. Technology Stack

This module is built inside Synapse rather than as a standalone service, since it already treats the Synapse application as a native intake channel and depends on Synapse's master data for identity resolution. Phase 1 requires no new stack decisions beyond one integration.

Layer	Choice	Note
Framework	Next.js 16 (App Router)	Matches Synapse exactly
Database	PostgreSQL via Prisma	Same database as Synapse — no new datastore
Authentication	Existing Synapse session authentication	Reused as-is; no new auth system
UI	Tailwind CSS 4	Already in use
Reporting / charts	Recharts	Already in use; covers the Phase 1 stats report and the Phase 2 dashboard
New for Phase 1	WhatsApp Business Cloud API, or a provider such as Gupshup / Twilio	The only new integration required for Phase 1, needed for the WhatsApp channel
New for Phase 3	Google Workspace API (Gmail)	For email intake and auto-categorisation
New for Phase 3	An LLM API for the after-hours bot	Retrieval over the curated solutions repository; keyword/category matching may be sufficient before any semantic-search investment is needed

12. Glossary

Term	Meaning
Ticket	A single tracked record of an issue, from intake through resolution.
Guest ticket	A ticket raised by someone not found in Synapse master data; captured with name and contact only, reconciled later.
Service request	A lighter-weight, automatable request — currently scoped to document delivery — distinct from a general support ticket.
Change request	A ticket reclassified as a development requirement and approved by management for the backlog, rather than resolved as a normal issue.
Solutions repository	A curated, human-approved set of past resolutions used to power the Phase 3 after-hours bot.

Master data	Synapse's existing <code>userdetails</code> / <code>hierarchy</code> tables, used to identify ticket raisers automatically.
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